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Order of execution

distinct and order by does not work together

Aggregation

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Group By

- total sales
- max order amount
- min order amount
- avg order value

input rows are 10000 rows

output rows are just 1

```
SELECT count(*) AS total_orders,  
SUM(amount) AS total_sales,  
MAX(amount) AS max_order_amount,  
MIN(amount) AS min_order_amount,  
AVG(amount) AS avg_order_value  
FROM sales;
```

count, sum, min, max, avg

```
SELECT * FROM sales;
```

```
SELECT 'total_orders' as aggregation, count(*) AS value FROM sales  
UNION  
SELECT 'total_sales', SUM(amount) AS total_sales FROM sales  
UNION  
SELECT 'max_order_amount', MAX(amount) AS max_order_amount FROM sales  
UNION  
SELECT 'min_order_amount', MIN(amount) AS min_order_amount FROM sales  
UNION  
SELECT 'avg_order_value', AVG(amount) AS avg_order_value FROM sales;
```

```
SELECT 'total_orders' as aggregation, count(*) AS value FROM sales  
UNION ALL  
SELECT 'total_sales', SUM(amount) AS total_sales FROM sales  
UNION ALL  
SELECT 'max_order_amount', MAX(amount) AS max_order_amount FROM sales  
UNION ALL  
SELECT 'min_order_amount', MIN(amount) AS min_order_amount FROM sales  
UNION ALL  
SELECT 'avg_order_value', AVG(amount) AS avg_order_value FROM sales  
UNION ALL
```

```
SELECT 'total_sales', SUM(amount) AS total_sales FROM sales;
```

CAN WE APPLY AGGREGATE FUNCTIONS ON DATE

SUM WILL NOT WORK FOR DATE , VARCHAR

AVG WONT WORK FOR DATE , VARCHAR

COUNT WILL WORK WITH EVERYTHING

MIN AND MAX(THESE WILL WORK WITH NUMERIC , STRING , DATE)

```
SELECT count(*) AS total_orders,  
SUM(amount) AS total_sales,  
MAX(amount) AS max_order_amount,  
MIN(amount) AS min_order_amount,  
AVG(amount) AS avg_order_value,  
MIN(sale_date) as min_sale_date,  
MAX(sale_date) as max_sale_date  
FROM sales;
```

Scripts :

```
CREATE TABLE IF NOT EXISTS sales (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    product VARCHAR(50),  
    category VARCHAR(50),  
    amount DECIMAL(10, 2),  
    sale_date DATE,  
    quantity INT,  
    customer_id INT,  
    store_location VARCHAR(50)  
);
```